Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Ethylene (C2H4)
Basis set   : cc-pVDZ
Charge   : 0
Multiplicity : 1
Total e-   : 16
n-body order : 3

GROUND STATE ENERGIES

E0 (Ha) : -78.05755506
E1 (Ha) : -77.89240700
Gap (Ha) : 0.16514806
HF (Ha) : -78.03837969

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 8

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(8,5)

Frozen core orbs : 4
Frozen e-   : 8
Active e-   : 8
Active orbs   : 5
Spin-orbitals   : 10
Hilbert dim   : 25

SPIN PROPERTIES

<S2>   : 0.000000
S   : 0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 4.000000
<D> / site : 0.800000

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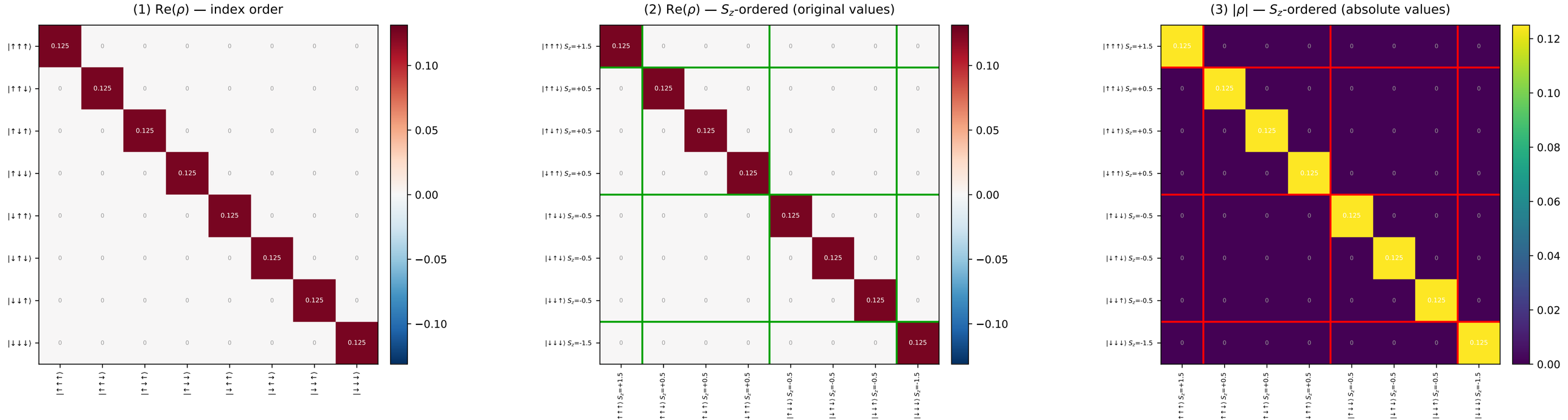
Body order : 3
Clusters   : 10

Date: 2026-07-15 09:33

Summary — 3-Body RDMs (Ethylene, CAS(8,5))

Orbitals	Purity	Entropy	C_GME (pure)	C_LB	C_gcg	C_UB	State
0,1,2	0.125000	2.079442	—	0.000000	0.001038	0.012754	STRONGLY MIXED
0,1,3	0.125000	2.079442	—	0.000000	0.001088	0.000000	STRONGLY MIXED
0,1,4	0.125000	2.079442	—	0.000000	0.000512	0.000000	STRONGLY MIXED
0,2,3	0.125000	2.079442	—	0.000000	0.001670	0.000000	STRONGLY MIXED
0,2,4	0.125000	2.079442	—	0.000000	0.000070	0.000000	STRONGLY MIXED
0,3,4	0.125000	2.079442	—	0.000000	0.000084	0.000000	STRONGLY MIXED
1,2,3	0.125000	2.079442	—	0.000000	0.000000	0.000000	STRONGLY MIXED
1,2,4	0.125000	2.079442	—	0.000000	0.001894	0.000000	STRONGLY MIXED
1,3,4	0.125000	2.079442	—	0.000000	0.000000	0.000000	STRONGLY MIXED
2,3,4	0.125000	2.079442	—	0.000000	0.004938	0.000000	STRONGLY MIXED

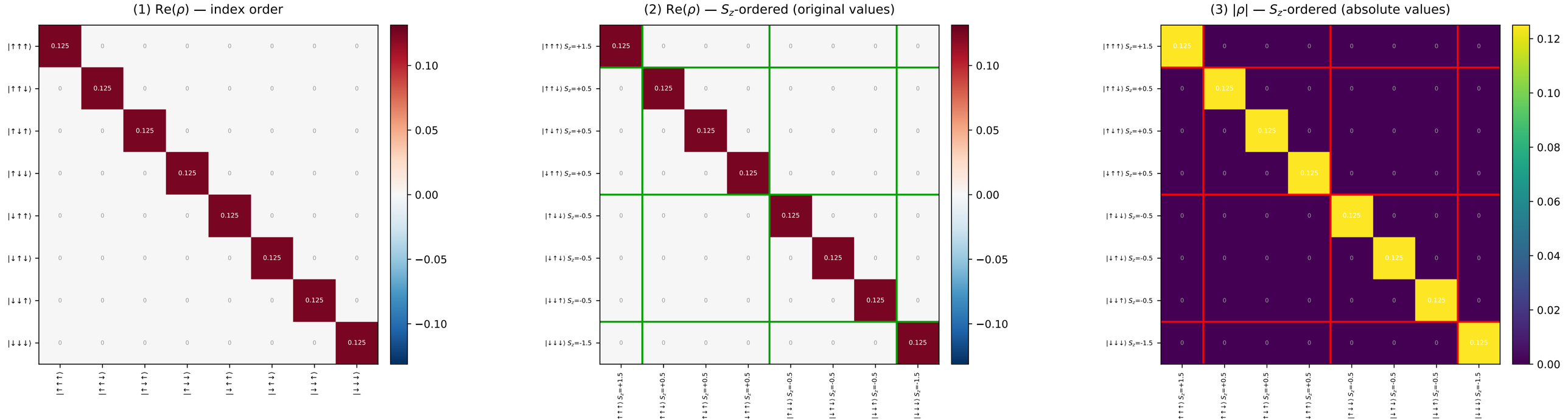
3-Body RDM (Sites 0,1,2) — same matrix in all three views



Trace=1.000000 Purity=0.125000 Entropy=2.079442 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.001038 ≤ C_{UB} = 0.012754 verdict: GME

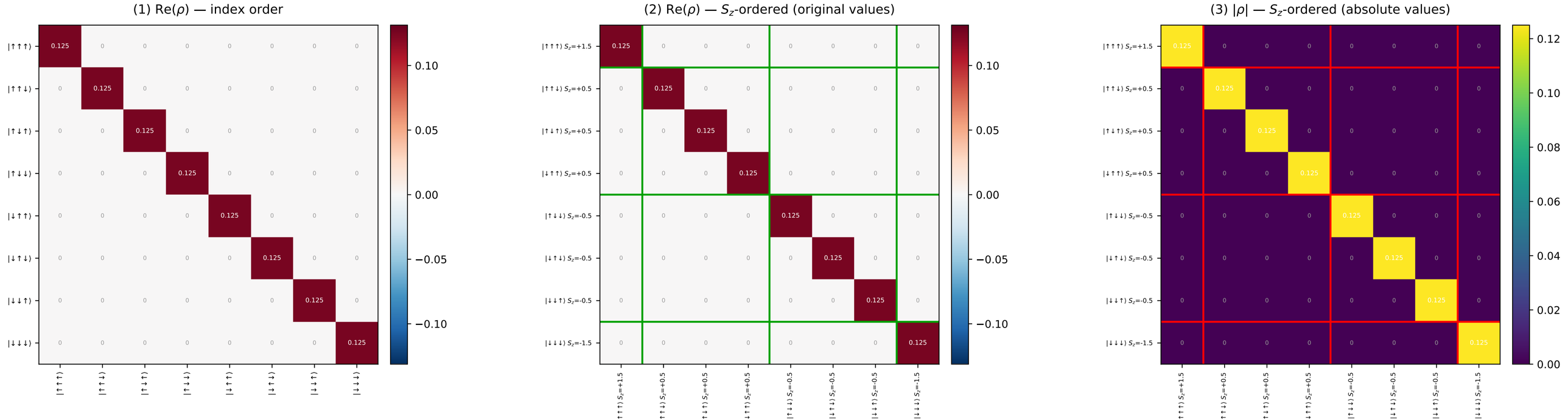
3-Body RDM (Sites 0,1,3) — same matrix in all three views



Trace=1.000000 Purity=0.125000 Entropy=2.079442 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.001088 ≤ C_{UB} = 0.000000 verdict: GME

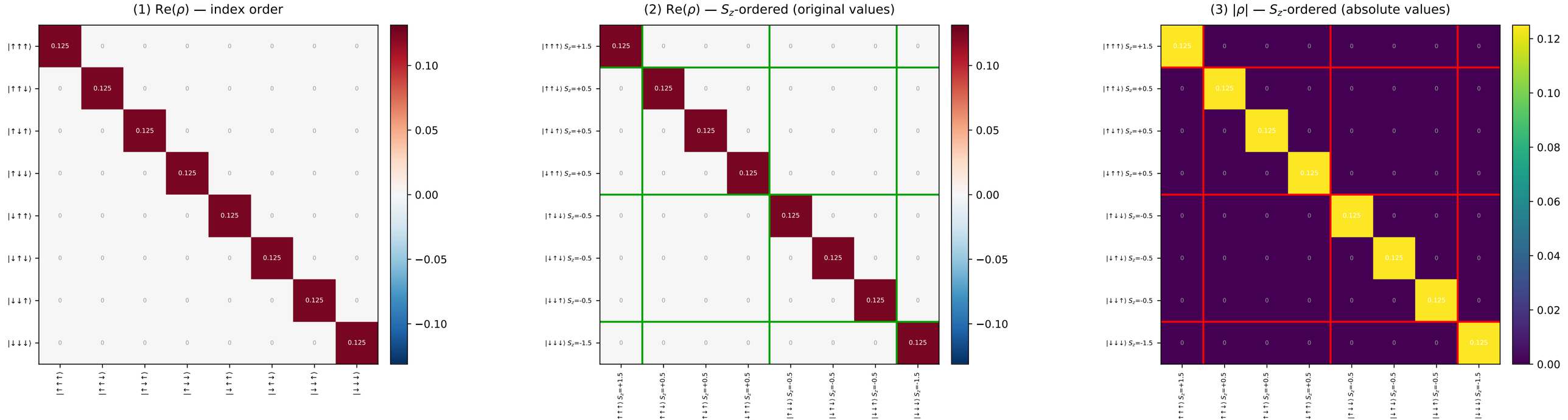
3-Body RDM (Sites 0,1,4) — same matrix in all three views



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GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000512 ≤ C_{UB} = 0.000000 verdict: GME

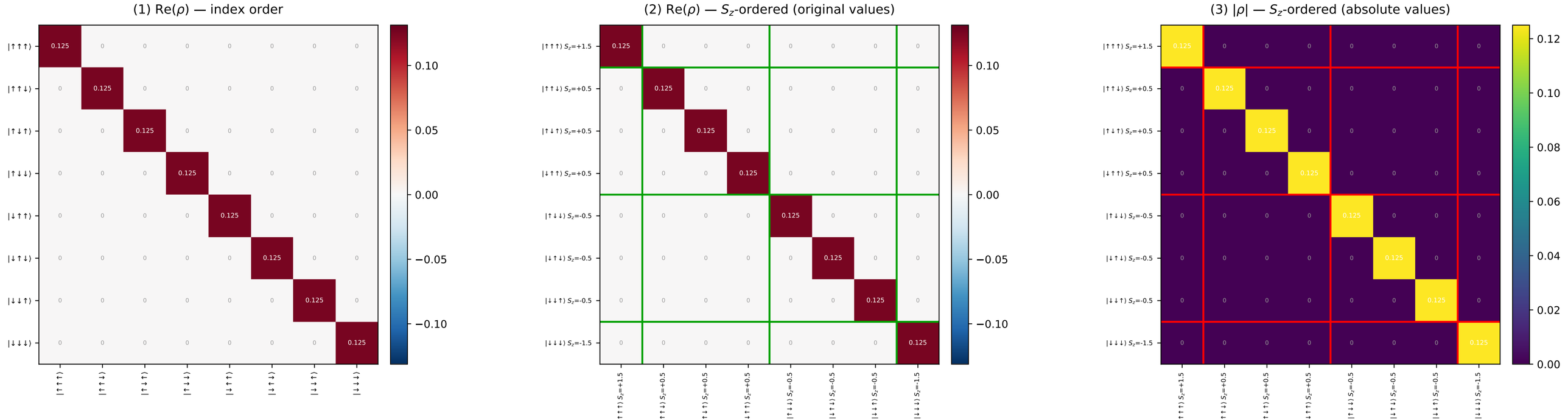
3-Body RDM (Sites 0,2,3) — same matrix in all three views



Trace=1.000000 Purity=0.125000 Entropy=2.079442 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.001670 ≤ C_{UB} = 0.000000 verdict: GME

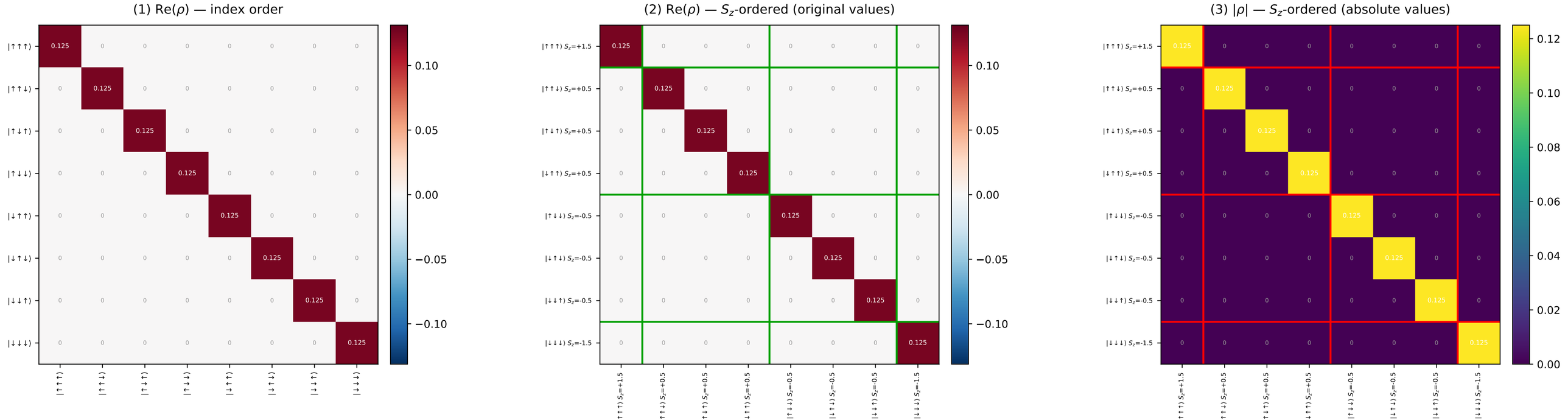
3-Body RDM (Sites 0,2,4) — same matrix in all three views



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GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000070 ≤ C_{UB} = 0.000000 verdict: GME

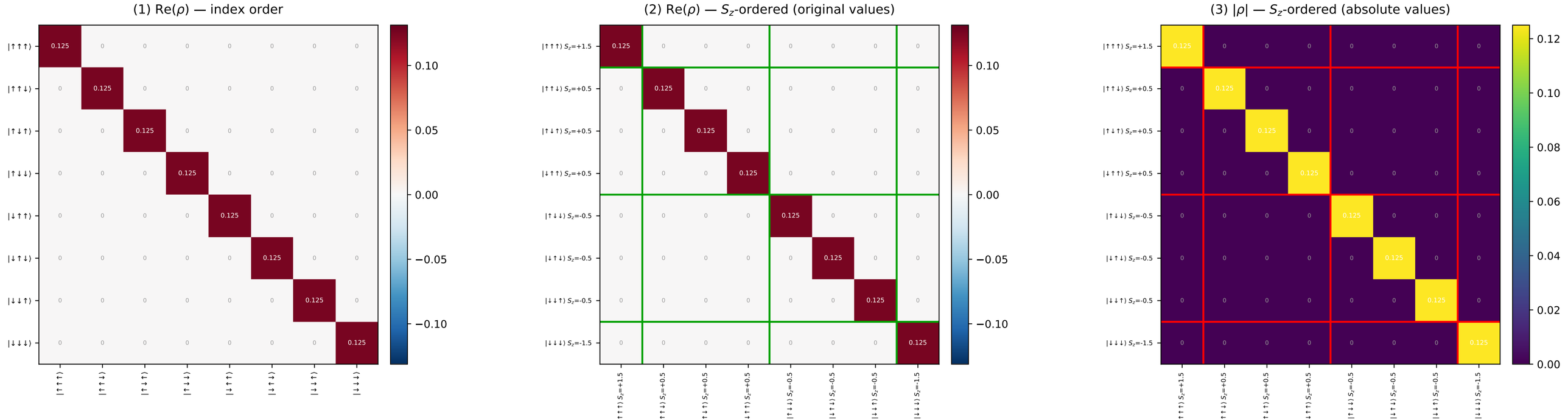
3-Body RDM (Sites 0,3,4) — same matrix in all three views



Trace=1.000000 Purity=0.125000 Entropy=2.079442 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000084 ≤ C_{UB} = 0.000000 verdict: GME

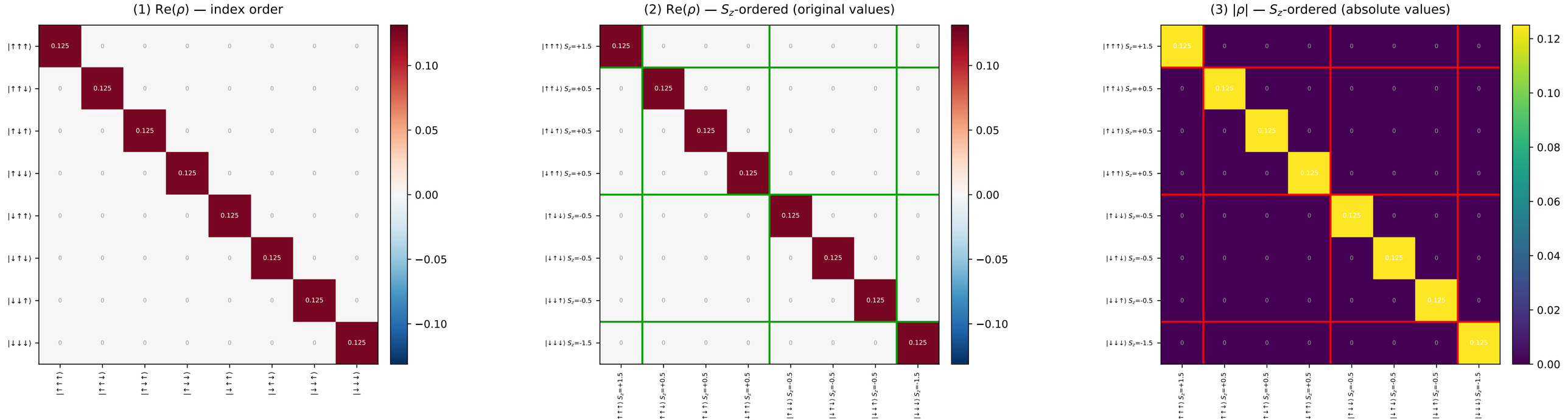
3-Body RDM (Sites 1,2,3) — same matrix in all three views



Trace=1.000000 Purity=0.125000 Entropy=2.079442 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000000 ≤ C_{UB} = 0.000000 verdict: no GME

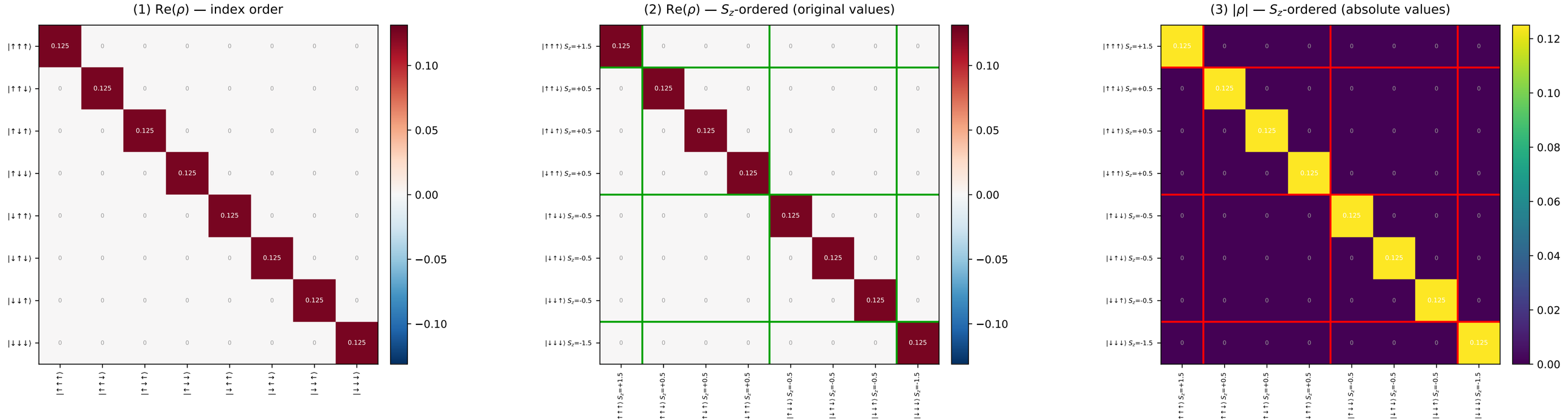
3-Body RDM (Sites 1,2,4) — same matrix in all three views



Trace=1.000000 Purity=0.125000 Entropy=2.079442 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.001894 ≤ C_{UB} = 0.000000 verdict: GME

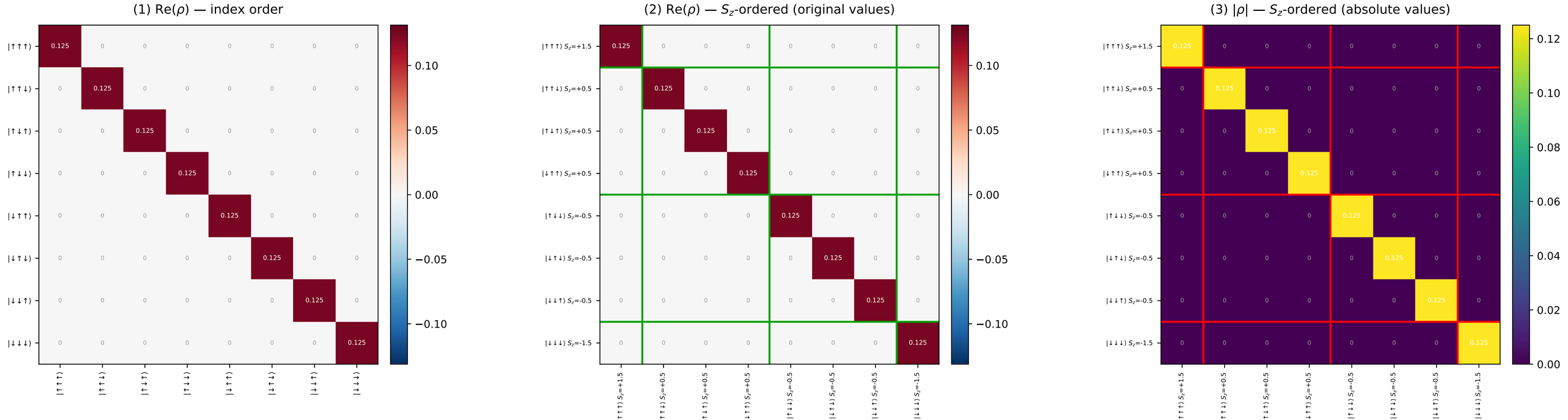
3-Body RDM (Sites 1,3,4) — same matrix in all three views



Trace=1.000000 Purity=0.125000 Entropy=2.079442 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000000 ≤ C_{UB} = 0.000000 verdict: no GME

3-Body RDM (Sites 2,3,4) — same matrix in all three views



Trace=1.000000 Purity=0.125000 Entropy=2.079442 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.004938 ≤ C_{UB} = 0.000000 verdict: GME